T01: what the torsion/frame cycle settles

PROVED in the supplied source system: $T = R_2 - R_1$ has infinite order. **TC-O1 is resolved. T01 remains OPEN.**

R_1 and R_2 are the positive-Y marked points of the exact two-width, six-scalar attachment. The fixed target is $P(x,y) = (x-y)^3+x+y$. We still seek arbitrarily large finite real A and a fixed $\varepsilon > 0$ with $|P(A,A)| \le |A|^{2-\varepsilon}$, counting every ordered pair of the same set.

The geometric distinction

Genus counts handles of a smooth compact complex curve. A **cover** is a finite map; a **branch point** is where its sheets merge. An **isogeny** is a finite-degree elliptic group homomorphism, unbranched in characteristic zero.

PROVED / TC-G1-G2. The genus-one sharing equation $u^3-v^3+u-v = c$ supplies one common column. A complete frame with the same ordered outputs also requires $3d^2+\ell^2 = 4$ and $2d^3-4d+c = 0$, where $d = u-v$ and $\ell = u+v$. The cubic has at most two admissible positive widths: at most four marked points, paired by column exchange.

PROVED / TC-G4. Completing one row requires $W_{cov}^2 = 4-3u^2$. Six simple branch points give genus four; a degree-e isogeny pullback gives genus $1+3e$. Riemann-Hurwitz excludes the stated rational elliptic completion. This does not exclude isolated real lifts, and that completion need not keep the second output.

The finite attachment and the whole square

PROVED / TC-G5: $N = 6$ scalars, $Q = 8$ controlled pairs, $Z = 2$ controlled outputs, $B = N^2-Q = 28$. Hence $|P(A,A)| \le Z+B = 30$. **EXPERIMENTAL / TC-X1, exact finite audit:** the actual image is 30, with two size-4 fibers and 28 singleton fibers. Two rich fibers do not control the remaining pairs.

Why the arithmetic certificate covers every order

A **valuation** v_{23} counts powers of 23. **Good reduction** means the integral curve stays nonsingular modulo 23. A **formal logarithm** turns group addition near the identity into ordinary addition, detecting nonzero points there as non-torsion.

$$T+\sigma(T) = 2R_2,$$
$$v_{23}(Y(2R_2)) = 1,\ v_{23}(X(4R_2)) = -2.$$

PROVED / MS-C1. With $a = \sqrt{2}$, $H = \sqrt{((3+a)/14)}$, $W = \sqrt{(4a-3)}$, all roots positive, $L = Q(a,H,W)$ has degree eight. The automorphism σ fixes a,H and negates W. The embedding $Q(a,H) \to Q_{23}$ with $a \equiv 18$, $H \equiv 6$ has good reduction. The supplied local logarithm lemma makes R_2 infinite order; the conjugation identity then does the same for T.

Meeting conclusion. Infinite order gives infinitely many distinct real elliptic multiples, not infinitely many admissible frames: the fixed-output locus still has at most four points. The marked torsion question, fixed-output width classification and stated rational elliptic completion obstruction are settled in scope. The cubic saving remains open. Sources: TC-G1/G2/G4/G5, TC-X1, MS-C1 and its saved referee appendix. This is a documentation freeze, with no new search, proof audit, experiment or next task.